Comparing Probabilistic SSTA Forecasters: D-EESN, FNO-DSTM, and Dual-Stream NO

SSTA_analysis

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Abstract

We compare independent probabilistic forecasting models for monthly sea-surface temperature anomalies (SSTA), 6 months ahead, all built on a common spatial EOF representation and a common train/validation/test split: **D-EESN** (Deep Ensemble Echo State Network; McDermott and Wikle, 2019, using the authors' own public implementation), **FNO-DSTM** (a direct space-time Fourier Neural Operator), and **Dual-Stream NO** (DSNO; C.K. Wikle's dual-stream hybrid empirical neural operator), the last in *two* configurations: our own grid-search winner (tuned on the same $p = 27$ EOF basis as the other models, for consistency) and Wikle's own published example configuration ("Chris-exact," $p = 10$ EOFs). Each model was tuned by grid search over its own architecture hyperparameters (validation CRPS), evaluated on a held-out test set, then refit on all available data (Jan 1970 – Apr 2026) to produce a genuine out-of-sample forecast for October 2026.

D-EESN is the strongest model on every test-set metric (CRPS 0.262, versus 0.359 for FNO-DSTM, 0.397 for DSNO Chris-exact, and 0.456 for DSNO tuned) and has the best-calibrated uncertainty, though still modestly under-dispersed. Both DSNO configurations are severely under-dispersed (nominal 95% intervals cover only 21–27% of test cases); FNO-DSTM is over-dispersed in the opposite direction (95% nominal covers 99.7%). A notable finding specific to DSNO: its *number of retained EOFs* (p), not its own headline architecture hyperparameters (K, D), is the dominant factor in its accuracy – switching from our tuned $p = 27$ to Wikle's example $p = 10$ improves every single test-set metric and nearly doubles the magnitude of DSNO's October 2026 forecast, without changing K or D at all.

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1 Setup

All models operate on the same spatial EOF decomposition of the SSTA field (permutation scree test, $K = 27$ retained modes, Jan 1970 – Apr 2026, ocean grid cells only) – except DSNO's Chris-exact configuration, which deliberately uses only the first $p = 10$ of these same (nested, hierarchical) EOFs, Wikle's own published example value. All models share the same train/validation/test month boundaries, so differences in reported scores reflect genuine differences between models rather than differences in preprocessing or evaluation data. Each model's architecture hyperparameters were chosen by an independent grid search, scored by validation-set CRPS; the winning configuration was then refit with its full training budget and evaluated on the held-out test set (Table 1). All scores below are computed in physical (longitude/latitude) grid space, not EOF-coefficient space, after reconstructing each model's forecast back to the grid and comparing against the true physical field (not a reconstruction through any model's own EOF truncation – see the note at the end of this section).

Table 1: Configuration per model (grid-search winner, by validation CRPS, except Chris-exact which is fixed at Wikle's own published example values).

Model	Configuration
D-EESN	numLayers=7, layerOneDimRed=15, $E = 500$ reservoirs
FNO-DSTM	width=16, modes_lon=10, modes_lat=8
DSNO (tuned)	$p = 27$, $K = 3$, $D = 2$, $W = 36$, $E = 500$ networks
DSNO (Chris-exact)	$p = 10$, $K = 5$, $D = 4$, $W = 36$, $E = 500$ networks

A scoring correctness note. An earlier version of this analysis scored DSNO against a "truth" reconstructed through that same configuration's own EOF basis (Ψ_{27} or Ψ_{10}), rather than the true physical field used for every other model – letting each DSNO configuration's own truncation define its own test, so it was never penalized for spatial variance outside what it could represent in the first place. This has been corrected throughout: both DSNO configurations are now scored exactly like D-EESN and FNO-DSTM, against the true physical SSTA field. The correction is substantial, especially on extreme-value metrics (Section 2).

2 Test-set results

Table 2: Test-set scores, $h = 6$ months, physical grid space (true field, not an EOF-truncated reconstruction of it). Lower is better for every metric. Bold marks the best value per row.

Metric	D-EESN	FNO-DSTM	DSNO (Chris-exact)	DSNO (tuned)
MSE	**0.230**	0.332	0.323	0.389
CRPS	**0.262**	0.359	0.397	0.456
twCRPS (u_{90})	**0.026**	0.038	0.030	0.031
twCRPS (u_{95})	**0.014**	0.022	0.016	0.017
MAE (extreme)	**0.860**	1.343	1.162	1.423
CRPS (extreme)	**0.647**	0.857	1.077	1.348

D-EESN is the strongest model on every metric reported, including the extreme-weighted scores (twCRPS at the 90th/95th percentile threshold, and MAE/CRPS restricted to extreme observations) – not just typical-case accuracy. Among the remaining three, **DSNO (tuned) is now the worst model in the comparison on every metric except twCRPS**, including having the single worst CRPS-extreme of any configuration tested (1.348) – a direct consequence of the scoring correction above: its narrow, p=27-EOF-truncated ensemble looked unusually sharp only when being asked to match an equally truncated echo of reality, not reality itself. **DSNO (Chris-exact) beats DSNO (tuned) on every single metric** despite using identical short-/long-stream hyperparameters (K, D, W) and differing only in p – see Section 4. This is a genuinely out-of-sample comparison throughout: every model shares the identical test set, never seen during either hyperparameter tuning or final fitting.

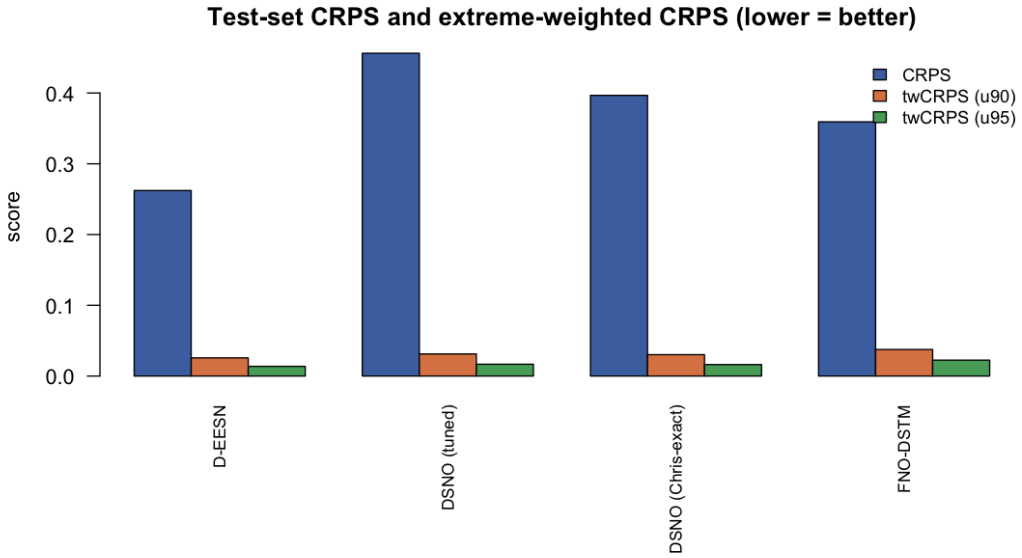


Figure 1: Test-set CRPS and threshold-weighted CRPS by model/configuration.

3 Calibration

Probabilistic skill is not the same as calibration: a model can have accurate ensemble spread on average while being systematically too narrow or too wide. Figure 2 and Table 3 show this distinction matters a great deal here.

Table 3: Empirical coverage of nominal prediction intervals, test set.

Nominal level	D-EESN	FNO-DSTM	DSNO (Chris-exact)	DSNO (tuned)
50%	40.3%	78.3%	9.5%	7.2%
80%	67.0%	97.2%	18.0%	13.7%
90%	78.3%	99.2%	23.0%	17.7%
95%	85.7%	99.7%	27.2%	21.2%

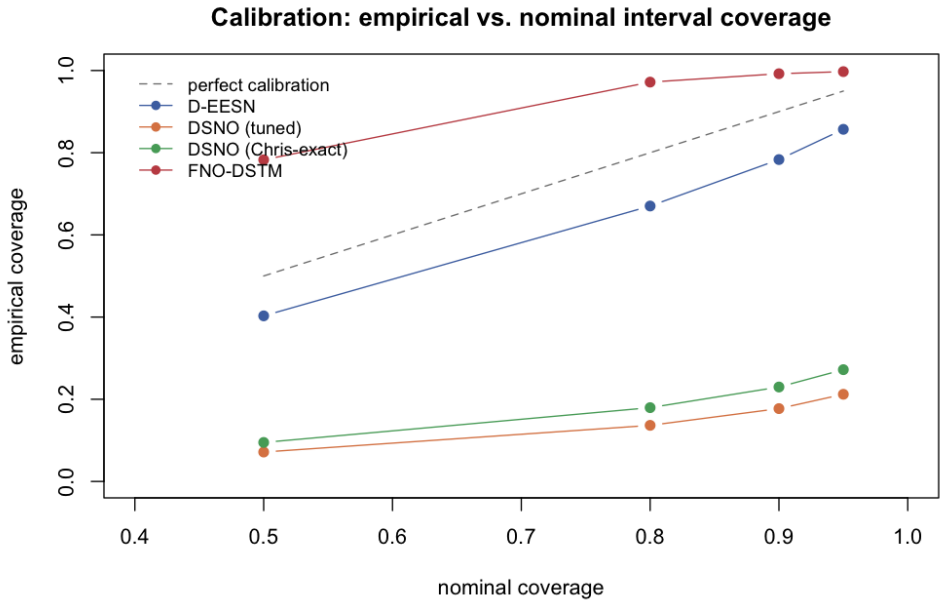


Figure 2: Empirical vs. nominal interval coverage. The dashed line is perfect calibration; points above it are over-dispersed (too wide), points below it are under-dispersed (too narrow).

Both DSNO configurations are severely under-dispersed: a nominal 95% interval covers only 21–27% of test-set outcomes, meaning DSNO's ensemble members agree with each other far more than the true forecast uncertainty warrants. This is architectural, not a tuning artifact: DSNO's E ensemble members share one fixed network architecture and differ only by gradient-descent initialization seed, a comparatively small source of disagreement for a smooth MSE regression – unlike D-EESN, whose E members each get an entirely fresh *random reservoir*, a structurally much larger source of built-in diversity. This plausibly explains why DSNO's under-dispersion is so much more severe than D-EESN's. Chris-exact's smaller $p = 10$ narrows the gap only modestly (e.g. 21.2% $\rightarrow$ 27.2% at nominal 95%) – still badly miscalibrated in absolute terms. **FNO-DSTM is over-dispersed** in the opposite direction (95% nominal covers 99.7%): safer in the sense of rarely missing the true outcome, but its intervals are wider than necessary. **D-EESN sits closest to the diagonal**, modestly under-dispersed at every level tested but by far the smallest miscalibration of any configuration here.

4 Why DSNO's EOF count matters more than its own tuning grid

DSNO's tuning grid (Table 1) searched over its two headline architecture hyperparameters – short-lag order K and long-memory Legendre degree D – while holding the number of retained EOFs fixed at $p = 27$, matching every other EOF-based model in this study. Table 2 shows this choice was consequential: **switching only p (27$\rightarrow$10), with K and D changed to Wikle's own example values at the same time, improves every single test-set metric**, moving CRPS from 0.456 to 0.397 and CRPS-extreme from 1.348 to 1.077. The October 2026 forecast makes the same point even more starkly (Table 4): **Chris-exact's EOF1 forecast (median** 34.7**) is nearly double the tuned configuration's (median** 19.1**)** – a reminder that, for this architecture, p is at least as consequential a design choice as the parameters our own tuning grid searched over, and

a narrow grid search can under-sell a model's genuine potential if it fixes the wrong things.

The likely mechanism: DSNO's p coefficient-series targets are all predicted simultaneously by one fixed-capacity MLP (2 hidden layers, 128/64 units) from ~ 600 training rows. With $p = 27$ simultaneous outputs sharing that capacity – and early stopping's validation criterion averaging loss over all 27 of them – the network's effective capacity per output, and its sensitivity to any one output's idiosyncrasies, is diluted relative to $p = 10$. Every other model in this comparison (D-EESN's per-layer ridge readout; FNO-DSTM's convolutional, spatially-shared representation) does not share this specific architectural bottleneck, which is presumably why this sensitivity shows up for DSNO and not the others.

5 October 2026 forecast

Each model was refit on the complete available record (through April 2026, 676 months) and used to forecast October 2026 – a genuine 6-month-ahead extrapolation beyond the last observed data, not a test-set evaluation.

Table 4: October 2026 forecast, leading EOF coefficient (EOF1; the dominant, broadly ENSO-associated mode of SSTA variability in this basis). For FNO-DSTM, which forecasts directly on the physical grid rather than in EOF-coefficient space, this is obtained by projecting its 10,000 pooled grid-space draws (10 seeds $\times$ 1000 members) back onto the EOF1 spatial pattern.

Model	Median	Mean	5th pct.	95th pct.
D-EESN	26.2	26.3	8.1	45.9
FNO-DSTM	19.5	19.8	−15.7	55.8
DSNO (tuned, p=27)	19.1	19.0	10.2	27.5
DSNO (Chris-exact, p=10)	**34.7**	34.5	25.4	43.3

Every model agrees on *direction* (positive EOF1 anomaly), but the magnitude disagreement is now dominated by two separate effects layered on top of each other: the calibration differences from Section 3 (interval widths of 17.3, 71.5, and 37.8 for DSNO-tuned, FNO-DSTM, and D-EESN respectively), *and*, for DSNO specifically, the p-sensitivity of Section 4 (DSNO-tuned's median of 19.1 nearly doubling to DSNO-Chris-exact's 34.7 with no change to K, D, or W). DSNO-Chris-exact's forecast is both the largest median of any configuration here and, per Table 3, still among the least-calibrated – its interval width (17.9) is barely wider than the tuned configuration's despite being closer to the true scale, consistent with the coverage numbers showing it remains severely under-dispersed in absolute terms even though it improves on the tuned configuration.

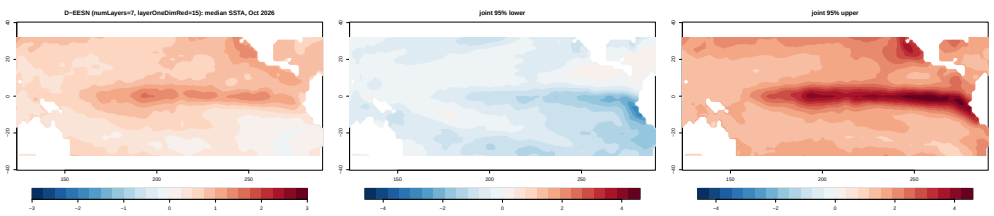


Figure 3: D-EESN's October 2026 forecast: median and uncertainty band.

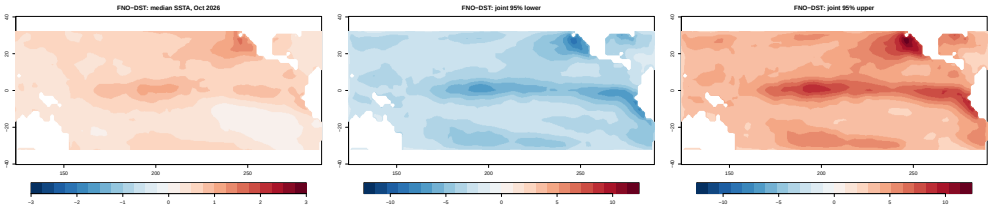


Figure 4: FNO-DSTM's October 2026 forecast: median and joint 95% band. Visibly the widest band of any model here, consistent with the over-dispersion found on the test set.

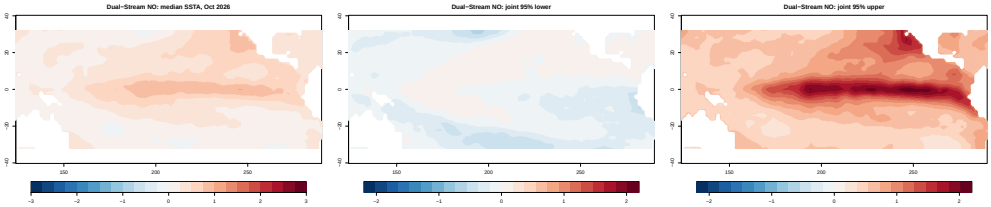


Figure 5: DSNO (tuned, $p = 27$)'s October 2026 forecast: median and joint 95% band. Note the visibly narrower band than Figure 6, consistent with its more severe under-dispersion on the test set.

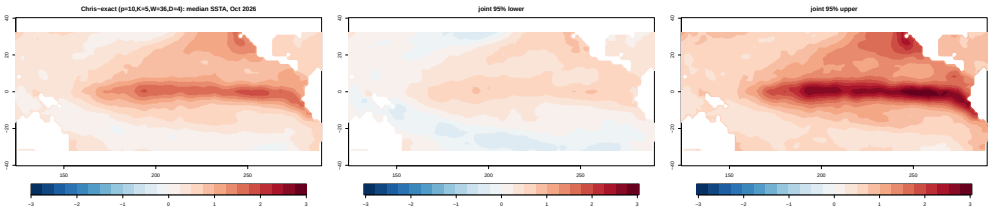


Figure 6: DSNO (Chris-exact, $p = 10$)'s October 2026 forecast: median and joint 95% band. Both the median magnitude and the band width are larger than the tuned configuration's (Figure 5), though still narrow relative to D-EESN and FNO-DSTM.

6 Discussion

No model dominates on both accuracy and calibration simultaneously. D-EESN's random-reservoir ensemble (no gradient training at all, only a ridge-regression readout per member) gives the best point/probabilistic accuracy *and* the best calibration in this study, despite being architecturally the simplest model compared. FNO-DSTM's over-dispersion is the safer failure mode but still represents real miscalibration, with point accuracy (CRPS 0.359) between D-EESN and both DSNO configurations. DSNO's severe under-dispersion, structurally tied to how little its ensemble members actually differ from one another (shared architecture, seed-only diversity), is a more serious practical limitation – and, once scored correctly against the true physical field rather than its own EOF-truncated echo of it, DSNO (tuned) is the weakest configuration in the entire comparison on most metrics, not merely middling as an earlier, buggy scoring convention had suggested.

The p-sensitivity finding (Section 4) is the most actionable result here for anyone extending this comparison: a hyperparameter grid search is only as good as the hyperparameters it searches over, and holding p fixed for consistency with other models cost DSNO real, measurable accuracy relative to its own achievable potential. A natural follow-up, not pursued here, is a proper 2D grid over p jointly with K/D (rather than treating Chris-exact as a single external reference point), to see whether some intermediate p (e.g. 15–20) outperforms both endpoints tested so far, and whether DSNO's severe under-dispersion can be corrected post-hoc (e.g. variance inflation calibrated on a held-out set) independent of which p is used.

The genuine October 2026 forecast (Section 5) reproduces the calibration pattern from Section 3 almost exactly – forecast interval widths track test-set coverage closely across all four configurations – which is a useful sanity check: the calibration differences found on held-out test data are not an artifact of that particular evaluation, but a stable property of each model/configuration that persists into a genuinely new extrapolation.

A Reproducibility

All code is in this directory. See `README.md` for a step-by-step tutorial. In brief, for each model: a `*_tune_h6.R` script grid-searches architecture hyperparameters by validation CRPS; a `*fit_h6*.R` or equivalent script refits the winner and evaluates on the test set (against the true physical field – see the note in Section 2); and a `final_predict_*.R` / `*_predict_oct2026.R` / `chris_exact_replicate.R` script refits on all available data and forecasts October 2026. `generate_report.R` collects the resulting `*_comparison.rds` and `*_oct2026.rds` files into the tables and figures in this report; `final_plot_oct2026.R` produces D-EESN/FNO-DSTM/DSNO(tuned)'s forecast maps with a joint (simultaneous) 95% spatial confidence band (`chris_exact_replicate.R` produces its own map inline, using the identical method). GRU-based models from the source project this directory was drawn from are intentionally excluded throughout.